A

Project Report On

"Underground Cable Fault Distance Detection System"

By

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CERTIFICATE

This is to certify that the project report titled **"Underground Cable Fault Distance Detection System "**has been carried out by **Dhiraj. D. Pardhi, Sumit. L. Bhange, Kunal. D. Pardhi, Maroti. B. Jadhav** under the guidance of **Dr. N. S. Lingayat.**

The work has been approved for the partial fulfillment of the during the academic session 2022-23 in Diploma Electrical Engineering of Dr.Babasaheb Ambedkar Technological University, Lonere.

Date:

Place: DBATU Lonere-Raigad.

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Head of Department

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Abstract

Underground cables are used for power applications where it is impractical, difficult, or dangerous to use the overhead lines. They are widely used in densely populated urban areas, in factories, and even to supply power from the overhead posts to the consumer premises. The underground cables have several advantages over the overhead lines; they have smaller voltage drops, low chances of developing faults and have low maintenance costs. However, they are more expensive to manufacture, and their cost may vary depending on the construction as well as the voltage rating. Underground cables are prone to a wide variety of faults due to underground conditions, wear and tear, rodents etc. Diagnosing the fault source is difficult and entire cable should be taken out from the ground to check and fix faults. The project work is intended to detect the location of fault in underground cable lines from the base station in km using Arduino UNO . To locate a fault in the cable, the cable must be tested for faults. This prototype uses the simple concept of Ohms law. The current would vary depending upon the length of fault of the cable. In the urban areas, the electrical cables run in underground instead of overhead lines. Whenever the fault occurs in underground cable it is difficult to detect the exact location of the fault for process of repairing that particular cable. The proposed system finds the exact location of the fault. The prototype is modelled with a set of resistors representing cable length in km and fault creation is made by a set of switches at every known distance to cross check the accuracy of the same. In case of fault, the voltage across series resistors changes accordingly, which is then fed to an ADC to develop precise digital data to a programmed Arduino that further displays fault location in distance. The fault occurring distance, phase, and time is displayed on 16x2 LCD interfaced with the Arduino.

ACKNOWLEDGEMENT

It is indeed a matter of great pleasure and proud privilege to be able to present this project report titled **"Underground Cable Fault Distance Detection System "**. The completion of the completion of the project work is a millstone in student life and its execution is inevitable in the hands of guide.

We are highly indebted the project the project guide **Dr. N. S. LINGAYAT** for his invaluable guidance and appreciation for giving form and substance to this report. It is due to his enduring efforts , patience and enthusiasm, which has given a sense of direction and purposefulness to this project and ultimately made it a success.

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CHAPTER-1

1.1 Introduction

In underground electricity distribution systems, the cables used are placed in the ground or in some form of ducts. This makes the cables strong and the chances of faults in them are very little. Whenever there is a fault in these cables, it becomes difficult to locate and repair the fault as conductors are not visible. Needless to say, detecting these faults is a lot like finding a needle in a haystack. There are many methods to locate the faults along with new detection technology and electrical items, which makes the task easier and less time consuming. However, do note that there is no single or a combination of methods to be considered as the "Best". There are different types of methods for different faults which make it safe and efficient to locate the faults without damaging the cable. Nevertheless, following are the electrical supply faults that occur in underground cables.

The purpose of this project is to determine the distance from the base station's underground cable fault in kilometers. In this project we used a simple concept of ohm's law. When a fault occurs in the system the distance located on liquid crystal display (LCD). Until the last decade, cables were designed to be placed above the head and, at present, there is no underground cable that is higher than the previous method. Adverse weather conditions such as storms, snow, torrential rains and pollution does not effect on underground lines, but when a fault occurs in underground lines it is difficult to locate the fault in underground cable. We will find the exact location of the fault. Now the world has become digitized so, the project is to detect exact location of the fault in digital form. Underground cabling system is a more common practice in many urban areas. Although the fault occurs for some reason, at that time, the repair process for this particular cable is difficult because of not knowing the exact location of the cable breakdown.

The light cable that we put inside the ground, through which we move the light from one place to another, this cable does not come out and we do not have any problem with these cables. But these cables have a problem. If there is a fault in them, then we do not know at which place the fault is. So we have a lot of trouble to find fault, this project has been made to solve this problem.

We use this project to detect underground cable fault. In this project, we have uploaded the program to Arduino, then Arduino will control the entire device. In this device, we have made a switch board to show the fault. In which resistance is also installed. All resistances have the same value. All the resistances are at equal distance. And one switch is controlling each resistance. And when all these resistances are applied Before that they have already been set and uploaded in distance programming. And the switches used in this circuit are used to cut power between resistances which will show how far the fault is and it will use three cables red line, blue line, yellow line. So when the switch is 2km in red line If we turn off the LCD will show 2km of fault, in this way we can find the fault for any line and at any distance

A bundle of electrical conductors used for carrying electricity is called as a cable. A under ground cable generally has one or more conductors covered with suitable insulation and a protective cover.

Commonly used materials for insulation are varnished cambric or impregnated paper. Fault in a cable can be any defect or non-homogeneity that diverts the path of current or affects the performance of the cable. So it is necessary to correct to correct the fault.

Power Transmission can be done in both overhead as well as in underground cables. But unlike underground cables the overhead cables have the drawback of being easily prone to the effects of rainfall, snow, thunder, lightning etc. This requires cables with reliability, increased safety, and ruggedness and greater services. So underground cables are preferred in many areas especially in urban places. When it is easy to detect and correct the faults in overhead line by mere observation, it is not possible to do so in an underground cable. As they are buried deep in the soil it is not easy to detect the abnormalities in them.

Even when a faults is found to be present it is very difficult to detect the exact location of the fault. This leads to debugging of the entire area to detect and correct the fault which in turn causes wastage of money and manpower. So it is necessary to know the exact location of faults in the underground cables. Whatever the fault is, the voltage of the cable has the tendency to change abruptly whenever a fault occurs. We make use of this voltage across change resistors to detect the fault.

In the event of short circuit (Line to Earth) fault, the voltage accordingly. It is then fed to an ADC to develop precise digital data that is directed to the programmed Arduino to display the same in kilometers. Hence this paper is very helpful for determining exact location of short circuit fault.

CHAPTER-2

PROPOSED MODEL WITH THEORETICAL BACKGROUND

2.1 Basic concept

The basic concept of Ohm"s law is found suitable in principle to develop a fault location tracking system. Based on the Ohm"s Law, it is found that the resistance of the cable is proportional to its length under constant conditions of temperature and the cross -section area and therefore if a low DC voltage is applied at the feeder end through a series of resistor in cable lines, the current would vary depending upon the location of fault in the cable. Here a system is developed which consists of a Arduino , LCD display, Fault Sensing Circuit Module and proper power supply arrangement. Hence, if there is a short circuit in the form of line to ground in any phase/phases, the voltage across series resistors changes accordingly and an analog signal in the form of voltage drop is generated by the fault sensing circuit of the introduced system, which is then fed to an ADC inbuilt in already programmed microcontroller to create the exact digital data and after processing the data the output will be displayed in the connected LCD with the exact location of fault occurred in kilometres from the source station and simultaneously also indicate the corresponding R, Y, B phase where fault occurred with the exact distance.

The circuit consists of a power supply, 4line display, Arduino and resistance measurement circuit. To induce faults manually in the kit, fault switches are used. About 12 fault switches are used which are arranged in three rows with each row having 4 switches. The 3 rows represent the 3 phases namely R,Y and B. The fault switches have 2 positions-No fault position (NF) and fault position(F). Main component of the underground cable fault detection circuit is low value resistance measurement. It is constructed using a constant current source of 100mAmps. It can measure very low value resistance as the cables have around 0.01 Ohm/meter resistance. For 10meter cable resistance becomes 0.1 Ohm. This circuit can measure resistance up 50 Ohm, Maximum cable length it can check up to 4 Kilometers.

So starting from the reference point 4 sets of resistances are placed in series. These 4 sets of resistances represent the three phases and the neutral. Short circuit faults, Symmetrical and unsymmetrical faults can be determined by this method. This project uses three set of resistances in series (ie)R10- R11-R12-R12,R17-R16-R14R21,R20-R19-R18-R25 one for each phase. Each series resistor represents the resistance of the underground cable for a particular distance and so here four resistances in series represent 1-4kms.Value of each resistance is 10kΩ.

2.2) Block diagram and circuit diagram

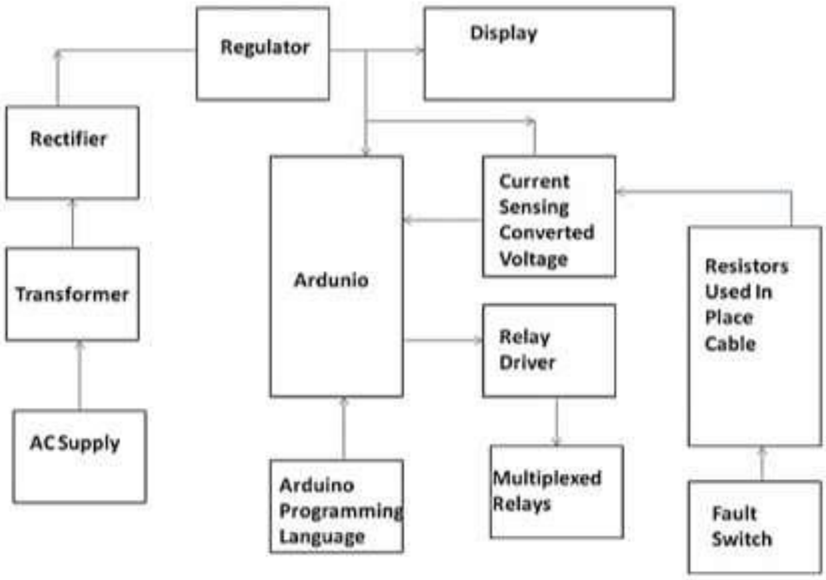

fig 2.2 Block diagram

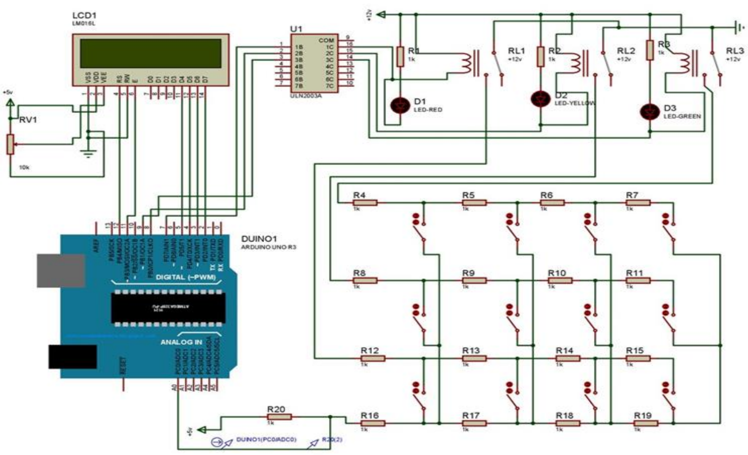

fig 2.2.1 circuit diagram

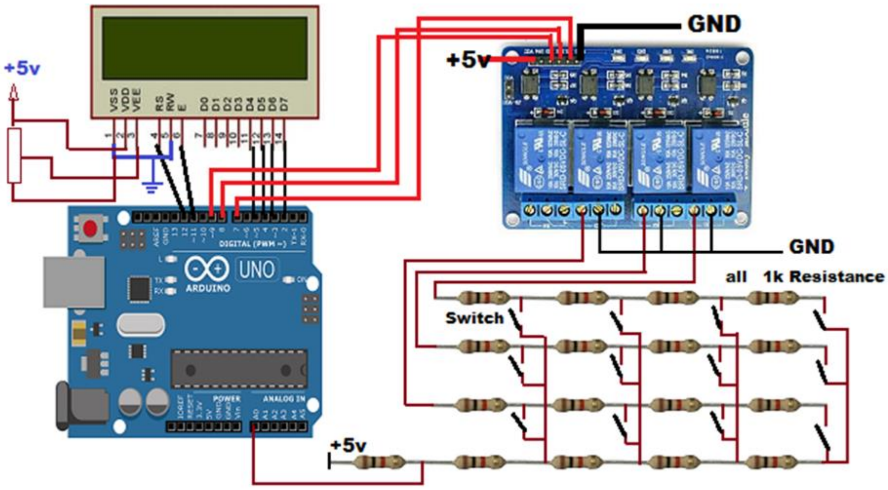

fig 2.2.2 circuit diagram

From the block diagram and Circuit fig 2.2, 2.2.1 and 2.2.2, A 230V AC supply is applied to the transformer from where it is stepped down to 12V AC. From the transformer the alternating current gets converted into direct current when it passes through a Bridge wave rectifier. The 12V DC then goes to the voltage regulator where it gets converted from 12V DC to 5V DC . Voltage regulator is used also converts the variable Dc supply into constant DC supply. This 5V DC is used to supply power to the arduino and the LCD. Power supply to the LCD is given from the voltage regulator.

The circuit consists of a power supply, 4 line display, arduino and resistance measurement circuit. To induce faults manually in the kit, fault switches are used. About 12 fault switches are used which are arranged in three rows with each row having 4 switches. The 3 rows represent the 3 phases namely R, Y and B. The fault switches: have 2 positions-No fault position (NF) and fault position (F).

So starting from the reference point 4 sets of resistances are placed in series. These 4 sets of resistances represent the three phases and the neutral. Short circuit faults, Symmetrical and unsymmetrical faults can be determined by this method. This project uses three set of resistances in series (ie) (R1-R2- R3-R4) ; (R5-R6-R7-R8) ; (R9-R10-R11-R12) one for each phase. Each series resistor represents the resistance of the underground cable for a particular distance and so here four resistances in series represent 1-4 kilometres. Value of each resistance is 1kΩ.but for showing long distance of cables they are multiplied by 2

One relay for each phase R, Y and B. three relays are used and the common points of the relays are grounded and the NO points are connected to the inputs of R4, R8 and R12 and being the three phase cable input. As supply needed for the relays is higher than that of the arduino, Relay driver is used to boost the supply and provide it to the relays.

When fault is induced by operating any of the 12 switches (to F position), they impose conditions like LG, LL, LLG fault as per the switch operation. As a result of the fault, there is a change in voltage value. This voltage value measured across the resistance is fed to the ADC of the Arduino. Using this value, the arduino computes the distance. Finally the distance of the fault from the base station is displayed in kilometers.

2.3) POWER SUPPLY

Power supply is given using AC adapter of 5V 2A because every component used in this project is work on 5V. if we supply more than or direct supply to this components that it will burn whole components due temperature rise to very hight temperature.

2.4) LCD Display

Liquid crystal display are interfacing to microcontroller 8051.Most commonly LCD used are 16*2 and20*2 display. In 16*2 display means 16 represents column and 2 represents rows. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden, such as preset words, digits, and 7-segment displays as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements.

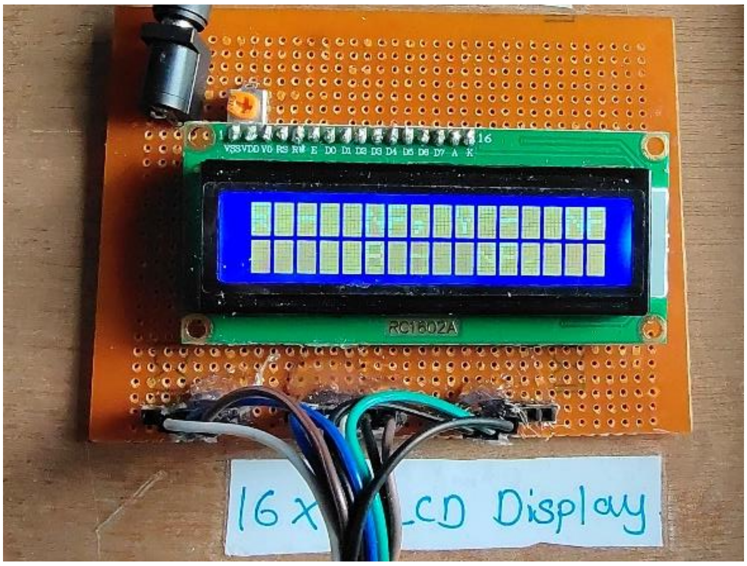

Fig 2.4 16x2 LCD Display

2.5) ARDUINO

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board.The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the

Fig 2.5 Arduino Uno

Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board – you can simply use a USB cable. Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program. Finally, Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible package.

2.6) Relay

Relay is sensing device which senses the fault and sends a trip signal to circuit breaker to isolate the faulty section. A relay is an automatic device by means of which an electrical circuit is indirectly controlled and is governed by change in the same or another electrical circuit. There are various types of relay: Numerical relay, Static relay and electromagnetic relay. Relay are housed in panel in the control room. Here three mini power relaya are used each for one of the three phases.The relays periodically scan the three ohases and send the signal to the arduino controller. The rating of each of the relays is about 12V.

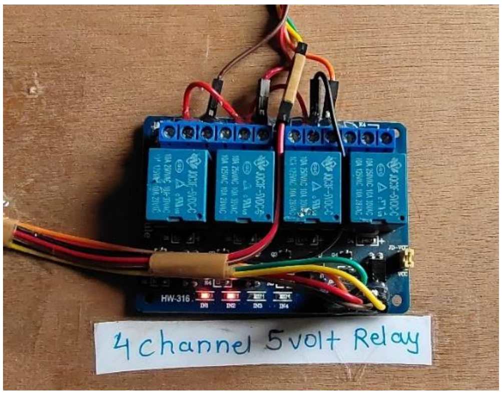

Fig 2.6 4 Channel 5 Volt Relay

2.7) Fault Sensing Circuit Module

The fault sensing circuit module is made by making use of a set of switches at every known equivalent kilometre as indicated by the set of series resistors to cross check accuracy of the same. The circuit uses four sets of resistors in series for each phase of the cable line i.e. R1, R2, R3, R4 for phase R, R5, R6, R7, R8 for phase Y and R9, R10, R11, R12 for phase B. Also resistors R13, R14 and R15 are used in series with supply line of each phase as shown in the circuit diagram. Each set of four series resistors represents the resistance of the underground cable for a specific distance of 4kms equally divided into 1km for each resistor. The resistors R13, R14 and R15 develop respective voltage drops corresponding to the occurrence of ground fault in one phase or two phases or three phases. This drop is then sensed by the ADC built in microcontroller. The other end of resistors R1, R3 and R5 are connected to ground.

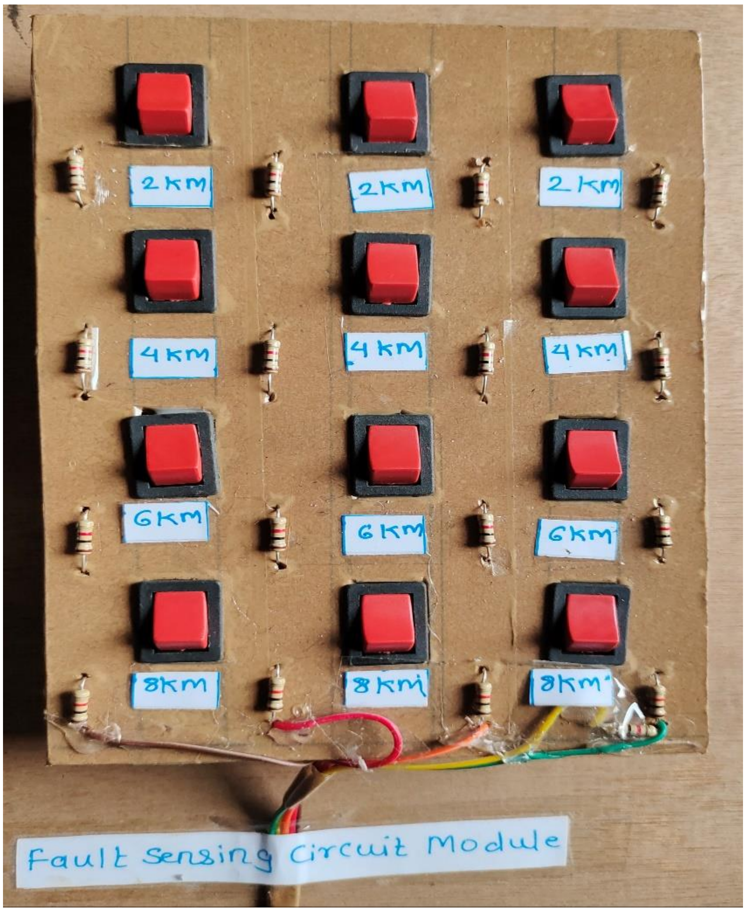

Fig 2.7 Fault Sensing Circuit Module

2.8 1K ohm Resistance:

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to **divide** voltages, bias active elements, and terminate transmission lines, among other uses.

1k0 / 1k ohm Resistor Colour Code

Value	**1 kΩ / 1000 Ω**
Type	4 Band Colour Code
Colour Code	Brown, Black, Red, Gold
Multiplier	Red, 100
Tolerance	Gold Band ±5%

CHAPTER-3

Arduino Uno

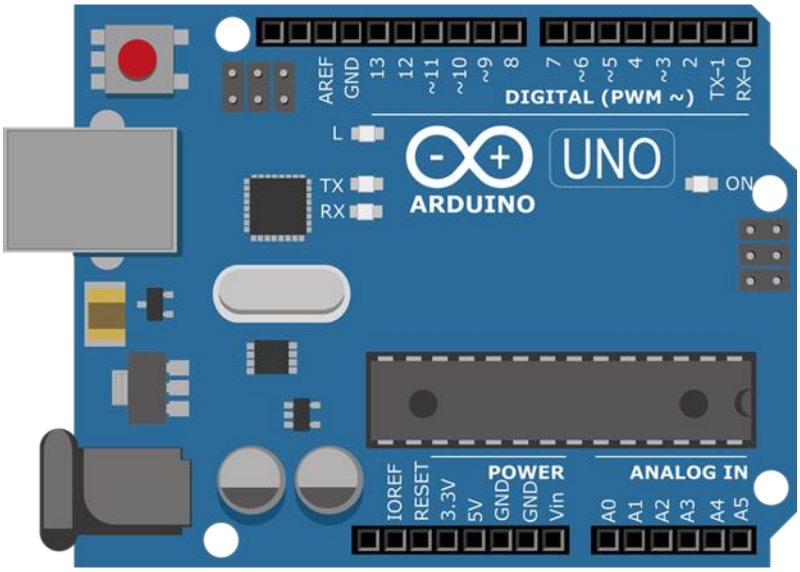

Fig 3 Arduino Uno

The Arduino UNO is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 Digital pins, 6 Analog pins, and programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable. It can be powered by a USB cable or by an external 9 volt battery, though it accepts voltages between 7 and 20 volts. The ATmega328 on the Arduino Uno comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer.

3.1 Technical specifications

- Microcontroller: Microchip ATmega328P
- Operating Voltage: 5 Volt
- Input Voltage: 7 to 20 Volts
- Digital I/O Pins: 14 (of which 6 provide PWM output)

- Analog Input Pins: 6
- DC Current per I/O Pin: 20 mA
- DC Current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB of which 0.5 KB used by bootloader
- SRAM: 2 KB
- EEPROM: 1 KB
- Clock Speed: 16 MHz
- Length: 68.6 mm
- Width: 53.4 mm
- Weight: 25 g

3.2 General Pin functions

- LED: There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.
- VIN: The input voltage to the Arduino board when it's using an external power source You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- 5V: This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 20V), the USB connector (5V), or the VIN pin of the board (7-20V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage the board.
- 3V3: A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- GND: Ground pins.
- IOREF: This pin on the Arduino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin

voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V.

- Reset: Typically used to add a reset button to shields which block the one on the board.

3.3 Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino board, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The 16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino Software (IDE) includes a serial monitor which allows simple textual data to be sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1). A Software Serial library allows serial communication on any of the Uno's digital pins.

CHAPTER-4
DESIGN PROCESS

4.1 Principle of Operation:

The system works mainly on the principle of Ohm's Law where a low DC voltage is applied at the feeder end through fault sensing circuit.

4.2 Hardware and Software Requirements:

4.2.1) Hardware Requirements: -

1. Resistors: Resistor is a passive component having two terminals. Resistors are connected in series which show the cables. We are using resistors of 10 kilo ohm in the project.
2. Switches: In our project we use switches to create fault in the circuit.
3. AC adapter: 5 Volt AC adapter is used to give the supply for whole circuit.
4. Microcontroller: The ATmega328P(Arduino) is a single-chip microcontroller created by Atmel. ATmega328 is commonly used in many projects and autonomous systems where a simple, low-powered, low-cost micro-controller is needed. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button.
5. Relay Driver**:** A Relay driver IC is an electro-magnetic switch that will be used whenever we want to use a low voltage circuit to switch a light bulb ON and OFF. Relays are switches that open and close circuits electromechanically or electronically.
6. LCD: LCD refers to Liquid Crystal Display, used in many devices (to display output). This is very convenient device that is used to display various information. We use a 16X2 LCD display.

4.2.2) Software Requirements:

1. Arduino Software (IDE): The publicly available Arduino Software (IDE) that provides a platform to write code in simple way and send to the microcontroller. This works on Linux, Mac OS and also on Windows.

Fig 4.2.2 Arduino Software (IDE)

4.3 Operation:

The operation of the system states that when the current flows through the fault sensing circuit module the current would vary depending upon the length of the cable from the place of fault that occurred if there is any short circuit fault with the Single Line to ground fault, or double line to ground fault, or three phase to ground fault.

The voltage drops across the series resistors changes accordingly and then the fault signal goes to internal ADC of the microcontroller to develop digital data. Then microcontroller will process the digital data and the output is being displayed in the LCD connected to the microcontroller in kilometres and phase as per the fault conditions. This output is also displayed in LCD display.

The power supply given to the system is 230V AC supply. This 230 V supply is fed to the Adapter Module. The adaptor module converts the AC voltage to DC. The ripple in output of adaptor module is then removed with the help of a 1000 microfarad electrolytic capacitor.

Since a constant 5V voltage source is desired for our system, because the Microcontroller (ATmega328), 16x2 LCD (Liquid Crystal Display), Relay Drivers and Relays, Fault Sensing Circuit Module, etc. and the other components work at 5V supply, hence we are using three voltage regulators . These voltage regulators convert the filtered output to 5V constant supply voltage.

The project consists of three relays The relays used here switches off/on the bulb loads R, Y and B to indicate the fault being occurred in corresponding phases.

4.4 Arduino Programming Code:

```
#include <LiquidCrystal.h>
// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

int sensorPin   = A0; // select the input pin for ldr
int sensorValue = 0; // variable to store the value coming from the sensor

void setup() {

  pinMode(7, OUTPUT);
    pinMode(8, OUTPUT);
      pinMode(9, OUTPUT);

  digitalWrite(7, LOW);
    digitalWrite(8, HIGH);
      digitalWrite(9, HIGH);

Serial.begin(9600); //sets serial port for communication

lcd.begin(16, 2);
  lcd.print("UNDERGROUND CABLE");
  lcd.setCursor(0, 1);
  lcd.print("FAULT DETECTOR");

  delay(1000);
  delay(1000);

}
void loop()
{
```

```
lcd.clear();
 digitalWrite(7, LOW);
   digitalWrite(8, HIGH);
     digitalWrite(9, HIGH);
      delay(350);

sensorValue = analogRead(sensorPin); // read the value from the sensor
Serial.println(sensorValue); //prints the values coming from the sensor on the screen

 if( (sensorValue >= 1000) )
{
    lcd.setCursor(0, 0);
     lcd.print("R -  NF,") ;
  Serial.print("R -  NF,") ;


 }


 else if( (sensorValue >= 890) && (sensorValue <= 920)  )
{
 Serial.print("R - 2KM,") ;
  lcd.setCursor(0, 0);
 lcd.print("R - 2KM,") ;
}
 else if( (sensorValue >= 860) && (sensorValue <= 880)  )
{
 Serial.print("R - 4KM,") ;
   lcd.setCursor(0, 0);
  lcd.print("R - 4KM,") ;
}
```

```
 else if( (sensorValue >= 800) && (sensorValue <= 825)  )
{
 Serial.print("R - 6KM,") ;
   lcd.setCursor(0, 0);
  lcd.print("R - 6KM,") ;
}
 else if( (sensorValue >= 670) && (sensorValue <= 688)  )
{
 Serial.print("R - 8KM,") ;
   lcd.setCursor(0, 0);
  lcd.print("R - 8KM,") ;
}
 delay(1500);

digitalWrite(7, HIGH);
   digitalWrite(8, LOW);
     digitalWrite(9, HIGH);
      delay(350);

sensorValue = analogRead(sensorPin); // read the value from the sensor
Serial.println(sensorValue); //prints the values coming from the sensor on
the screen

 if( (sensorValue >= 1000) )
{

  Serial.print("Y -  NF") ;
   lcd.setCursor(8, 0);
   lcd.print(" Y -  NF,") ;
 }

 else if( (sensorValue >= 890) && (sensorValue <= 920)  )
```

```
{
  lcd.setCursor(8, 0);
 Serial.print("Y - 2KM,") ;
  lcd.print(" Y - 2KM,") ;
}
  else if( (sensorValue >= 860) && (sensorValue <= 880)  )
{
 Serial.print("Y - 4KM,") ;
 lcd.setCursor(8, 0);
   lcd.print(" Y - 4KM,") ;
}
  else if( (sensorValue >= 800) && (sensorValue <= 825)  )
{
 Serial.print("Y - 6KM,") ;
 lcd.setCursor(8, 0);
   lcd.print(" Y - 6KM,") ;
}
  else if( (sensorValue >= 670) && (sensorValue <= 688)  )
{
 Serial.print("Y - 8KM,") ;
 lcd.setCursor(8, 0);
   lcd.print(" Y - 8KM,") ;
}
 delay(1500);

digitalWrite(7, HIGH);
    digitalWrite(8, HIGH);
      digitalWrite(9, LOW);
       delay(350);

sensorValue = analogRead(sensorPin); // read the value from the sensor
Serial.println(sensorValue); //prints the values coming from the sensor on
```

the screen

```
 if( (sensorValue >= 1000) )
{
  lcd.setCursor(5, 1);
   Serial.println("B -  NF,") ;
    lcd.print("B -  NF,") ;
 }

  else if( (sensorValue >= 890) && (sensorValue <= 920)  )
{
 Serial.println("B - 2KM,") ;
 lcd.setCursor(5, 1);
  lcd.print("B - 2KM,") ;
}
  else if( (sensorValue >= 860) && (sensorValue <= 880)  )
{
 Serial.println("B - 4KM,") ;
 lcd.setCursor(5, 1);
   lcd.print("B - 4KM,") ;
}
  else if( (sensorValue >= 800) && (sensorValue <= 825)  )
{
 Serial.println("B - 6KM,") ;
 lcd.setCursor(5, 1);
   lcd.print("B - 6KM,") ;
}
  else if( (sensorValue >= 670) && (sensorValue <= 688)  )
{
 Serial.println("B - 8KM,") ;
 lcd.setCursor(5, 1);
   lcd.print("B - 8KM,") ;
```

```
}
 delay(1500);

}
```

CHAPTER-5

AC ADAPTOR

An AC adapter, AC/DC adapter, or AC/DC converter is a type of external power supply, often enclosed in a case similar to and AC plug. Other common names include plug pack, plug-in adapter, adapter block, domestic mains adapter, line power adapter, wall wart, power brick, and power adapter.

Use of an external power supply allows portability of equipment powered either by mains or battery without the added bulk of internal power components, and makes it unnecessary to produce equipment for use only with a specified power source; the same device can be powered from 120 VAC or 230 VAC mains, vehicle or aircraft battery by using a different adapter. Another advantage of these designs can be increased safety; since the hazardous 120 or 240 volt mains power is transformed to a lower, safer voltage at the wall outlet and the appliance that is handled by the user is powered by this lower voltage.

5.1 Modes of operation:

Fig 5.1 AC adapter

An AC adapter disassembled to reveal a simple, unregulated linear DC supply circuit: a transformer, four diodes in a bridge rectifier, and an electrolytic capacitor to smooth the waveform Originally, most AC/DC adapters were linear power supplies, containing a transformer to convert the mains electricity voltage to a lower voltage, a rectifier to convert it to pulsating DC, and a filter to smooth the pulsating waveform to DC, with residual ripple variations small enough to leave the powered device unaffected. Size and weight of the device was largely determined by the transformer, which in turn was determined by the power output and mains frequency. The output voltage of these adapters varied with load; for equipment requiring a more stable voltage, linear voltage regulator circuitry was added. Losses in the transformer and the linear regulator were considerable; efficiency was relatively low, and significant power. contains a transformer operating at a high frequency and outputs direct current at the desired voltage. The high-frequency ripple is more easily filtered out than mains-frequency. The high frequency allows the transformer to be small, which reduces its losses; and the switching regulator can be much more efficient than a linear regulator. The result is a much more efficient, smaller, and lighter device. Safety is ensured, as in the older linear circuit, because a transformer still provides galvanic isolation. dissipated as heat even when not driving a load.

5.2 Advantage:

External AC adapters are widely used to power small or portable electronic devices. The advantages include:

- Safety – External power adapters can free product designers from worrying about some safety issues. Much of this style of equipment uses only voltages low enough not to be a safety hazard internally, although the power supply must out of necessity use dangerous mains voltage. If an external power supply is used the equipment need not be designed with concern for hazardous voltages inside the enclosure. This is particularly relevant for equipment with lightweight cases which may break and expose internal electrical parts.

- Heat reduction – Heat reduces reliability and longevity of electronic components, and can cause sensitive circuits to become inaccurate or malfunction. A separate power supply removes a source of heat from the apparatus.

- Electrical noise reduction – Because radiated electrical noise falls off with the square of the distance, it is to the manufacturer's advantage to convert potentially noisy AC line power or automotive power to "clean", filtered DC in an external adapter, at a safe distance from noise-sensitive circuitry.

- Weight and size reduction – Removing power components and the mains connection plug from equipment powered by rechargeable batteries reduces the weight and size which must be carried.

- Ease of replacement – Power supplies are more prone to failure than other circuitry due to their exposure to power spikes and their internal generation of waste heat. External power supplies can be replaced quickly by a user without the need to have the powered device repaired.

CHAPTER-6

LCD Display

6.1 LCD Display

We come across LCD displays everywhere around us. Computers, calculators, television sets, mobile phones, digital watches use some kind of display to display the time. An LCD is an electronic display module which uses liquid crystal to produce a visible image. The 16×2 LCD display is a very basic module commonly used in DIYs and circuits. The 16×2 translates o a display 16 characters per line in 2 such lines. In this LCD each character is displayed in a 5×7 pixel matrix.

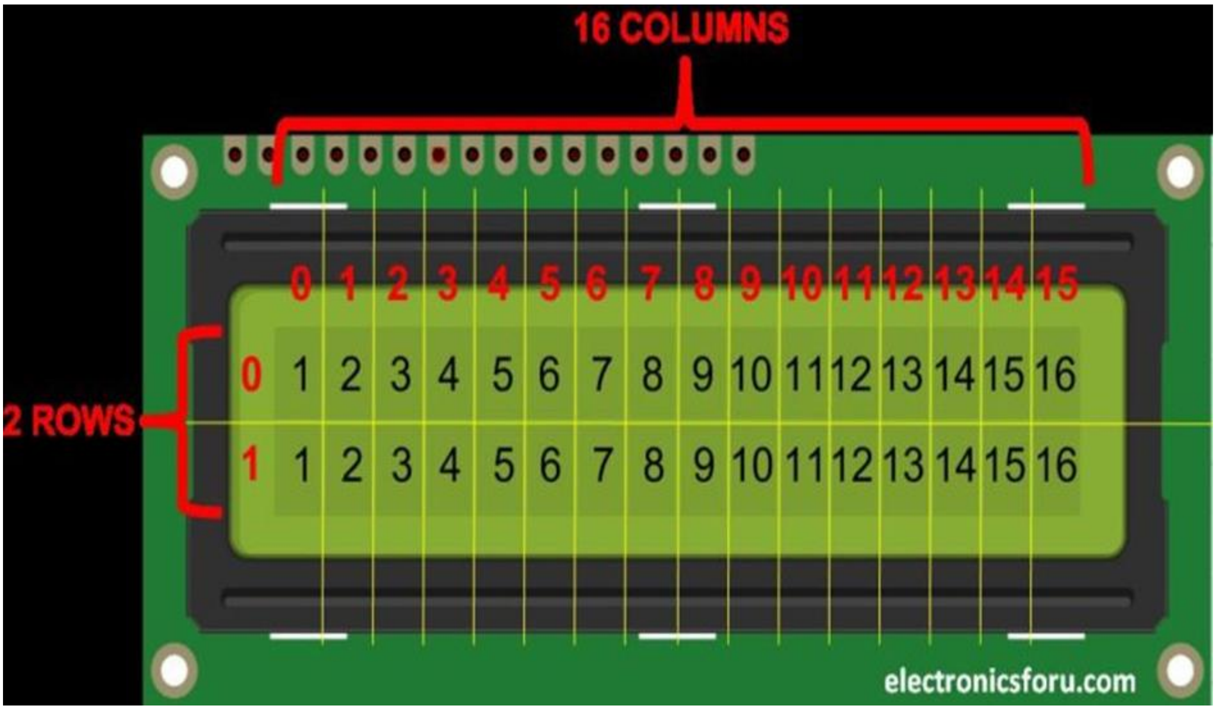

Fig 6.1 LCD Display

A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals. Liquid crystals do not emit light directly, instead using a backlight or reflector to produce images in color or monochrome. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden, such as present words, digits, and seven-segment displays, as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements. LCDs can either be normally on (positive) or off (negative), depending on the polarizer arrangement.

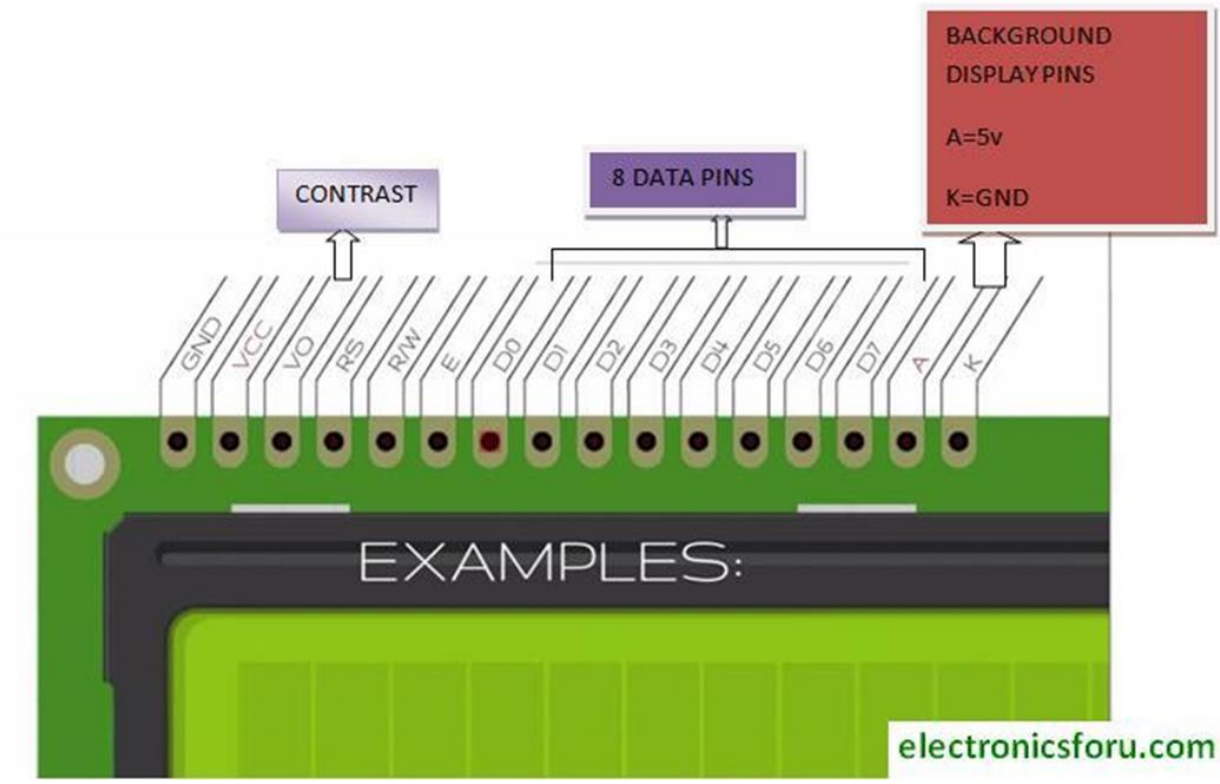

Fig 6.2 LCD Display Connection Pins

LCDs are used in a wide range of applications, including LCD televisions, computer monitors, instrument panels, aircraft cockpit displays, and indoor and outdoor signage. Small LCD screens are common in portable consumer devices such as digital cameras, watches, calculators, and mobile telephones, including smartphones. LCD screens are also used on consumer electronics products such as DVD players, video game devices and clocks. LCD screens have replaced heavy, bulky cathode ray tube (CRT) displays in nearly all applications. LCD screens are available in a wider range of screen sizes than CRT and plasma displays, with LCD screens available in sizes ranging from tiny digital watches to very large television receivers.

The LCD screen is more energy-efficient and can be disposed of more safely than a CRT can. Its low electrical power consumption enables it to be used in battery-powered electronic equipment more efficiently than CRTs can be.

CHAPTER-7
4 Channel 5 volt Relay Kit

7.1 Introduction

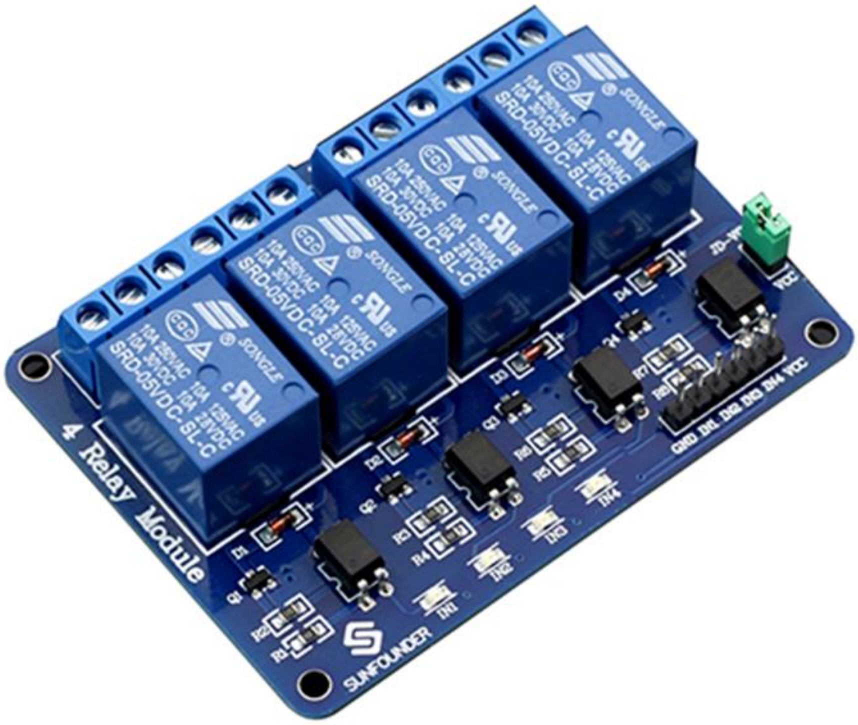

Fig 8.1 4 channel 5v relay kit

This is a 5V 4-channel relay interface board, and each channel needs a 15-20mA driver current. It can be used to control various appliances and equipment with large current. It is equiped with high-current relays that work under AC250V 10A or DC30V 10A. It has a standard interface that can be controlled directly by microcontroller.

7.2 Principle

From the picture below, you can see that when the signal port is at low level, the signal light will light up and the optocoupler 817c (it transforms electrical signals by light and can isolate input and output electrical signals) will conduct, and then the transistor will conduct, the relay coil will be electrified, and the normally open contact of the relay will be closed. When the signal port is at high level, the normally closed contact of the relay will be closed. So you can connect and disconnect the load by controlling the level of the control signal port.

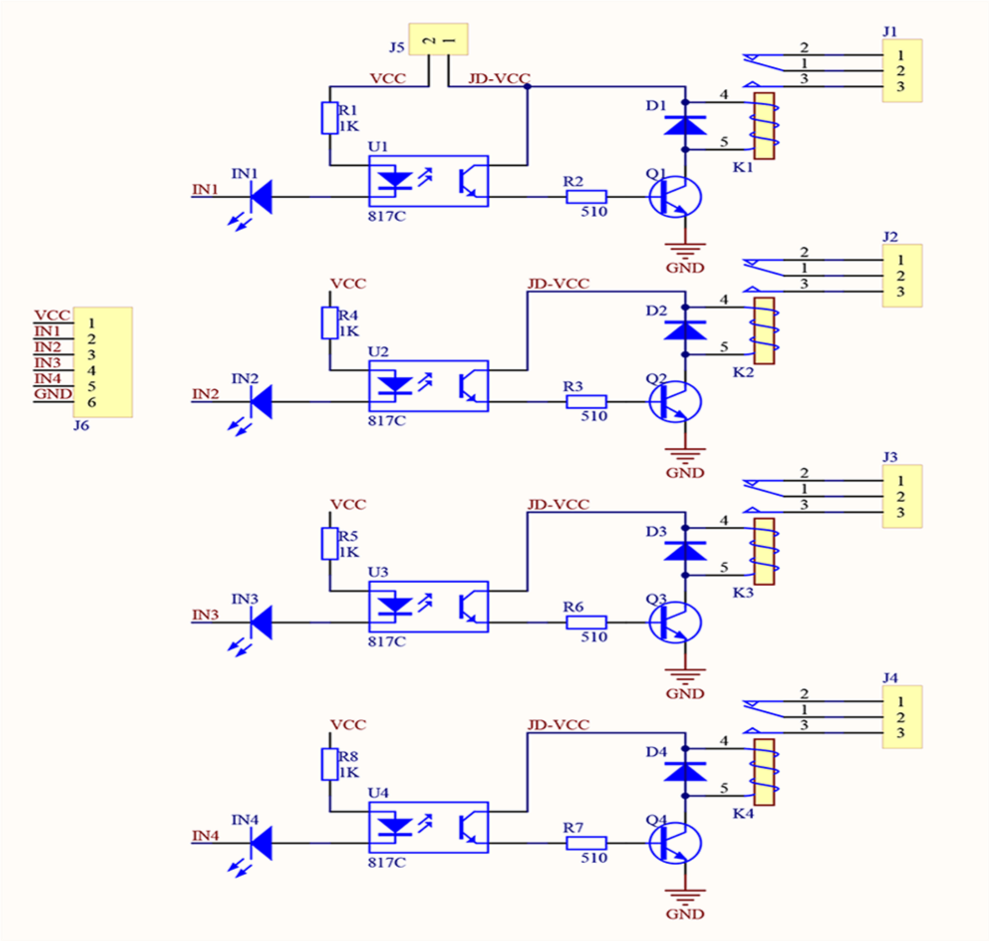

fig 7.2 4 channel 5v relay Circuit diagram

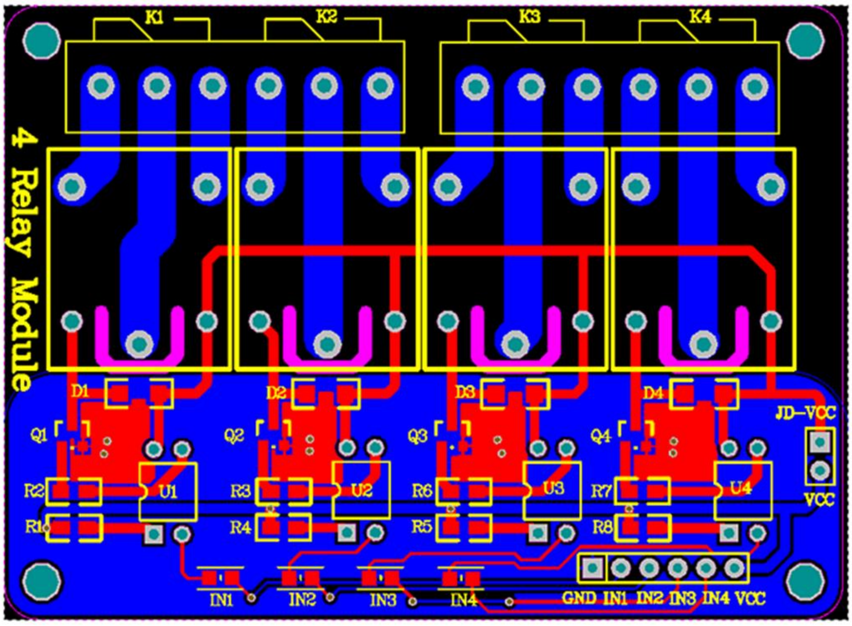

fig 7.2.1 channel 5v relay PCB layout

7.3 Pin Description:

Input:

VCC: Positive supply voltage

GND: Ground

IN1--IN4: Relay control port

Output:

Connect a load, DC 30V/10A， AC 250V/10A

- Features：

- Size: 75mm (Length) * 55mm (Width) * 19.3mm (Height)
- Weight: 61g
- PCB Color: Blue
- Closed at low level with indicator on, released at high level with indicator off
- VCC is system power source, and JD_VCC is relay power source. Ship 5V relay by default. Plug jumper cap to use
- The maximum output of the relay: DC 30V/10A, AC 250V/10A

CHAPTER-8

RESULTS AND DISCUSSIONS

8.1 Simulation Results

A fault detecting system is developed using microcontroller ATmega328P IC, LCD display, Relays, fault sensing circuit, etc.

In the simulation fault is created intentionally by closing the fault switch. The corresponding voltage, current and distance values are processed by the microcontroller and are displayed on the LCD.

8.2 Hardware Results:

The hardware model shown in fig 8.2 is meant for single phase supply. The power supply is provided using step-down transformer, rectifier, and regulator (AC adapter used). The current sensing circuit of the cable provides the magnitude of voltage drop across the resistors to the microcontroller and based on the voltage the fault distance is located.

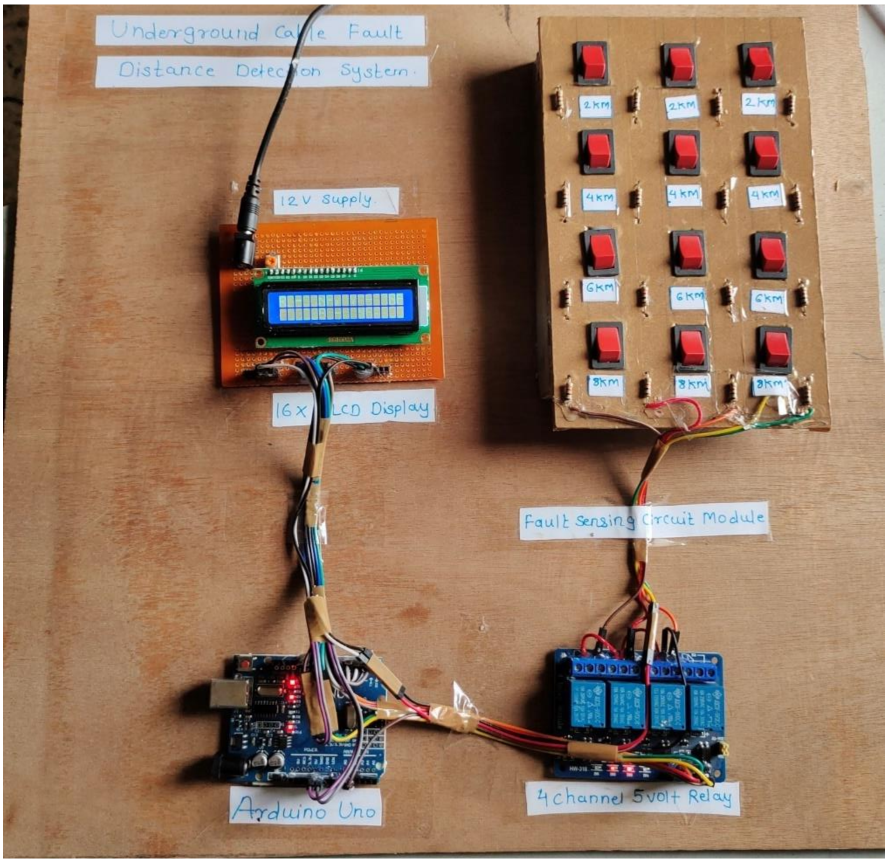

Fig 8.2 Hardware model Underground Cable Fault Distance Detection System

Figures 8.3,8.4,8.5, and 8.6 show the hardware results for fault locations at 6 Km & 8 Km respectively. The output is updated onto the LCD Display . the operator can monitor and determine the fault location from the base station .

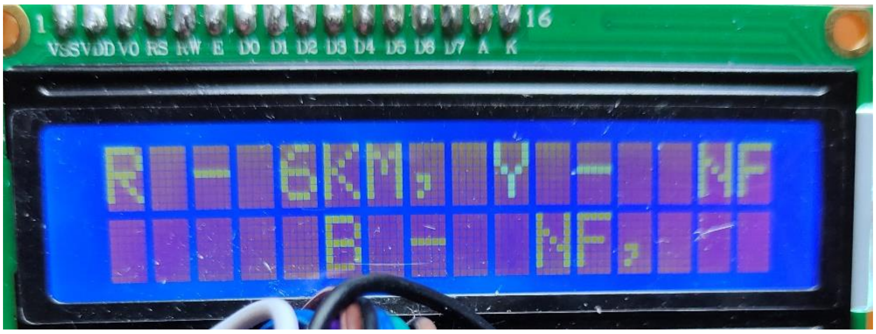

Fig 8.3 LCD displaying the fault at 6KM distance

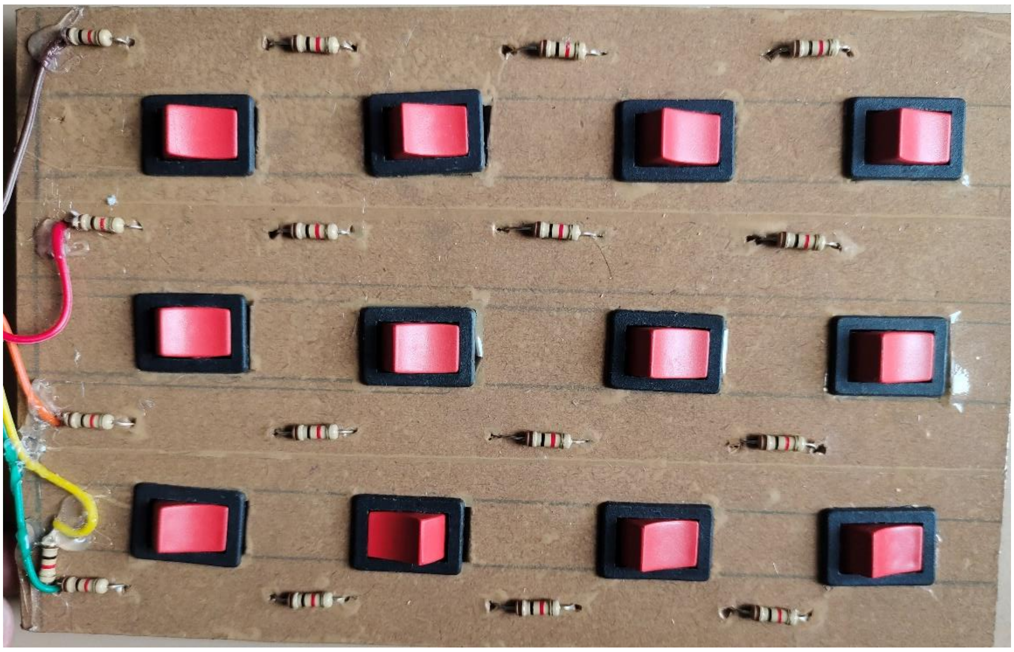

Fig 8.4 Creating Fault by closing the switch no.3 on R phase.

Fig 8.5 LCD displaying the fault at 8KM distance

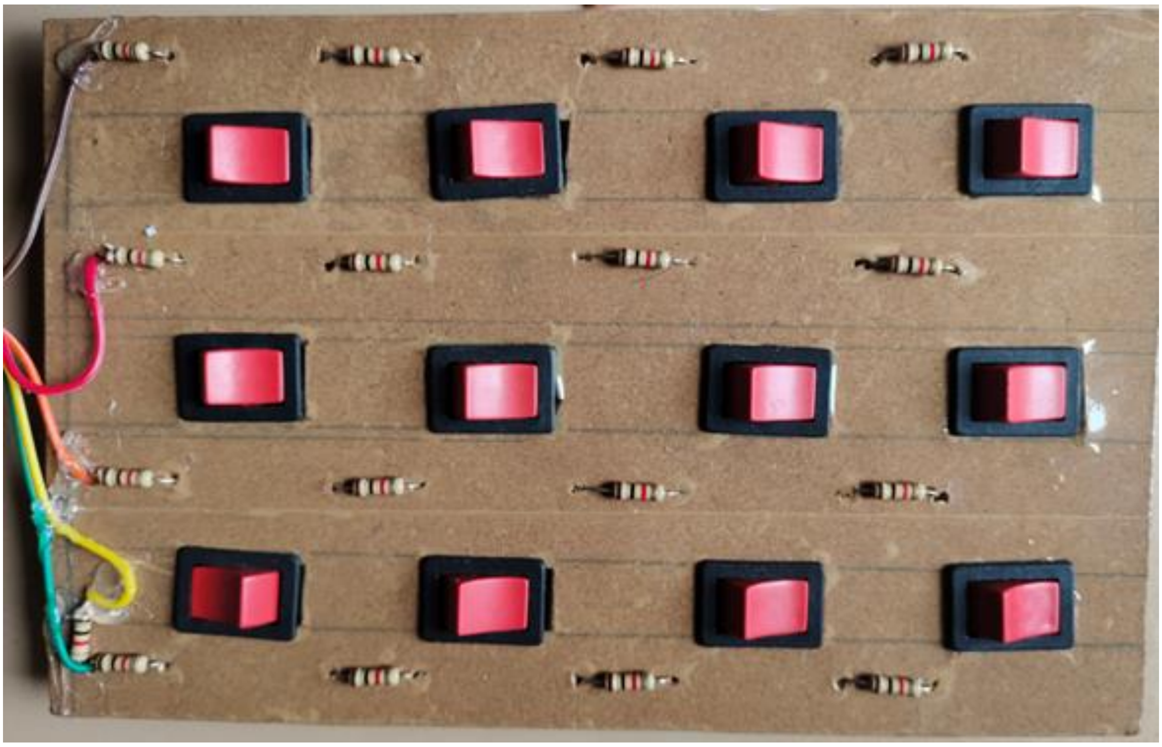

Fig 8.6 Creating Fault by closing the switch No.4 on R phase

CHAPTER 9
CONCLUSIONS

This project described the Underground Cable Fault Distance Detection System Using Arduino Uno in software and hardware simulation form and results were successful. A full-fledged prototype model had been implemented as a proof of concept to realize and understand the real time scenarios in underground cable system. Through this prototype simulation model the proposed architecture had been demonstrated that can effectively satisfy the requirement of exact fault location detection in the underground cable system and it is believed that this model can be a promising technology to solve future fault location detection problem.

With the help of " Underground cable Fault Detector" we are able to find the fault in the wire rather than removing the whole wire. It becomes very difficult to repair in case of any faults because finding the exact location of the fault in such cable system is quite difficult. With the proposed system, finding the exact location of the fault is possible and to repair it. It help in consuming time .

Reference

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